## Abstract

The aim of the project is to relate coalescence dynamics of bubbles to the presence of dissolved salts and surfactants in a liquid by modelling interface boundary conditions for the drainage of the thin liquid film between approaching bubbles. Bubble coalescence directly determines the size of bubbles in a bubbly flow. Bubble size is a critical parameter for the transport of chemical species in bubble column reactors. The fact that salt water limits coalescence of bubbles can be seen from the existence of froth in sea water and the absence of the same in fresh water. A number of experiments in literature report the ability of salts to limit the coalescence of bubbles, when the salt concentration exceeds a critical value. However, owing to limited theoretical and numerical studies, the molecular mechanisms behind control of bubble coalescence by certain salts is not well understood. Yet phenomenological models can be built based on experimental observations. In this project we will first focus on modelling interface boundary conditions for the drainage of the thin liquid film between approaching bubbles thereby relating stability of the thin films to salt concentration in the liquid. The interface model will be employed in a meshless particle based method called Smoothed Particle Hydrodynamics (SPH) or a mesoscopic variant of it in the length and time scale of the thin liquid films. Using a hybrid approach this film scale computations will then be coupled to the flow at the scale of the bubbles, in order to decide the stability of the thin liquid film and its possible rupture. Larger systems with hundreds of bubbles in three dimensions will be simulated using SPH at the bubble scale as well, in order to relate transport properties in the bulk of the flow to the presence and concentration of salts and surfactants through this proposed coupling procedure.



## Zusammenfassung

Ziel dieses Projekts ist es den Effekt von gelsten Salzen und Tensiden auf die dynamische Koaleszenz von Gasblasen innerhalb einer Flssigkeit zu untersuchen, indem Grenzflchenrandbedingungen fr das Abflieen des dnnen Films zwischen aufeinander zukommenden Blasen modelliert werden. Blasenkoaleszenz hat einen direkten Einfluss auf die Blasengre in Blasenstrmungen. Die Blasengre wiederum ist ein kritischer Parameter fr den Transport von chemischen Stoffen in Blasensulenreaktoren. Die Tatsache, dass gelstes Salz die Koaleszenz von Blasen behindern kann, spiegelt sich in der Bildung von Schaum auf Salzwasser und dessen Abwesenheit auf Swasser wieder. In der Literatur gibt es eine Reihe von Experimenten, die zeigen, dass Salze die Koaleszenz von Blasen behindern knnen, wenn die Salzkonzentration einen kritischen Wert berschreitet. Jedoch wurden zu diesem Thema bisher nur sehr wenige theoretische und numerische Untersuchungen durchgefhrt, weshalb der Mechanismus hinter der Kontrolle von Blasenkoaleszenz mithilfe von bestimmten Salzen noch nicht zur Gnze verstanden ist. Dennoch ist es mglich phnomenologische Modelle auf Basis experimenteller Untersuchungen zu entwickeln. Im ersten Abschnitt dieses Projekts liegt der Fokus auf der Modellierung von Grenzflchenrandbedingungen fr das Abflieen des dnnen Flssigkeitsfilms zwischen sich annhernden Blasen. Dadurch kann die Stabilitt dnner Filme in Abhngigkeit der Salzkonzentration untersucht werden. Das entwickelte Grenzflchenmodell wird in die gitterfreie, partikelbasierte Simulationsmethode Smoothed Particle Hydrodynamics (SPH) oder in eine mesoskopische Variante auf der Lngen- und Zeitskala der dnnen Flssigkeitsfilme eingebettet. Um ber die Stabilitt bzw. ein mgliches Aufreien des dnnen Flssigkeitsfilms entscheiden zu knnen, werden die Simulationen des Flssigkeitsfilms mit Hilfe eines hybriden Ansatzes mit der Strmung auf Lngenskala der Blasen gekoppelt. Grere Systeme mit hunderten von Blasen in drei Dimensionen werden ebenfalls mit SPH auf Lngenskala der Blasen simuliert, um die Transporteigenschaften innerhalb der Strmung mit der Anwesenheit und Konzentration von Salzen und Tensiden mit Hilfe der oben vorgeschlagenen Kopplung in Verbindung zu setzen.

# Project Description

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# Control of Bubble Coalescence

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# Project Description

## 1 State of the art and preliminary work

### 1.1 State of the art

**a) Thin Liquid Films**

Bubble coalescence is central to many industrial phenomenon such as foam fractionation, oil recovery from tar sands and floatation process for mineral recovery using air bubbles. Control of coalescence of bubbles is important in the above mentioned industrial phenomena as well as in vital biological phenomenon like decompression sickness.

The stability of foams and hence the coalescence of bubbles is related to the thinning and rupture of intervening film of liquid between two gas bubbles [38]. The process of formation, thinning and eventual rupture of this thin liquid film (TLF) is schematically shown in 1a. The first two phases, namely approach and formation of the 'dimple' until formation of the TLF is governed by hydrodynamics. The ensuing process involves draining of the TLF owing to thinning of the film due to capillary pressure for the initial part. Further thinning and rupture of the TLF is closely related to constituents such as surfactants and dissolved salts in the liquid.

Adsorption of surfactants at the film surface plays a critical role in stability of TLF and is exhaustively studied in literature [18, 42, 61, 66]. The effect of surfactants on the TLF's stability is well understood.

In addition to surfactants, the presence of salts is known to affect the stability of foam liquid films. The performance of TLFs under the influence of salts is studied experimentally [41, 55] and often these studies are individually limited in scope [39]. The experimental studies include monitoring bubble population in a bubble column [15, 58, 69], and bubble scale studies [28, 67] and using optical interferometry to study life of TLFs. Thus experimental studies have spanned macroscopic observations to microscopic information on the mechanism for stability of liquid films in the presence of salts. Most of these studies focus independently on the concentration of the electrolyte and on the type of electrolytes.

Despite extensive experimental studies, there are only few models in literature to theoretically describe this phenomenon [35, 60, 68]. Most of these models are derived from lubrication theory and try to match experimental observations by modifying the Hamacker constants. A comprehensive model that describes the stabilizing effects of salts is yet to be proposed. However based on separate arguments the stability of films is related to a set of causes such as ion specificity, hydrophobic attraction of interfaces, and affinity of certain ions to gas-liquid interfaces.

**b) Causes for inhibition of coalescence in presence of salts**

- **Concentration**

In [67] a quantitative study of effect of different salts and their concentration on pairs of coalescing bubbles was quantitatively studied, and a sharp decrease in coalescence was observed beyond different critical concentrations at different gas flow rate. A swarm of bubbles was studied in the same context in [58]. This and a relatively recent study [15] suggest an influence of gas flow rate on the transition concentration of the salt.

– **Types of salts**
Collins [51] studied water affinities of ions and showed that the effect of an ion on the molecular structure of water depends on charge density. This work also classified ions into 'kosmotropes' and 'chaotropes'. Following this Craig et. al [58] proposed a combining rule based on experimental observations, to classify the ion combinations that resulted in the inhibition of bubble coalescence. This classification is based on polarizability of the ion. This was further studied in [33] with the conclusion that salts with one ion having propensity for the interface and the other ion for the bulk of the liquid inhibited bubble coalescence.

– **Hydrophobic attraction and dissolved gases**
Hydrophobic interactions between the gas-liquid interfaces of approaching bubbles is known to play an important role in inhibition of bubble coalescence. Hydrophobic attractions are considerably stronger than van der Waals attraction forces [36, 52] and is used an a plausible mechanism that governs TLF stability. The electric double layer forces is also vanishingly small in the presence of dissolved salts.

Film drainage in a generic way was presented initially by Marrucci [68] using the pressure balance equation:

$$-\frac{4\Gamma}{h^2}\frac{d\sigma}{dC} - \frac{2\sigma}{R} + \Pi(h) = 0, \tag{0.1}$$

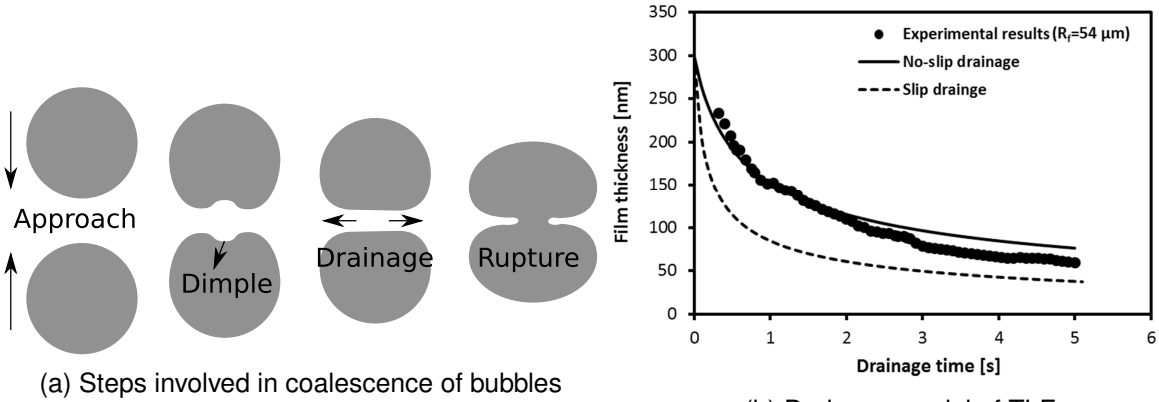
where the term $d\sigma/dC$ is the surface tension gradient with respect to the molar concentration $C$ of the dissolved salt, $h$ is the film thickness, $\Gamma$ is the Gibb's surface excess, and $2\sigma/R$ and $\Pi(h)$ are the capillary and disjoining pressures, respectively. The disjoining pressure can be written as a sum of van der Waals forces and hydrophobic attraction. Experimental studies show that the strength of hydrophobic attractions decrease considerably in "degassed" water relative to normal water between hydrophobized solid surfaces [13, 36, 47]. The long-ranged hydrophobic attractions in foam films is explained as a result of increased local disruption of hydrogen bonds of water molecules due to migration of dissolved gases. In essence, the migration of dissolved gases decreases the effective viscosity of water film, resulting in faster drainage due to increased surface mobility. Salts are known to reduce the solubility of gases and this phenomenon is described by the Setchenov equation [63]:

$$\log\left(\frac{S^\circ}{S}\right) = K_s \cdot C, \tag{0.2}$$

where $S^\circ$ and $S$ are the solubility of gas in pure water and saline solutions respectively. $K_s$ and $C$ are the salting effect coefficient and concentration of salt respectively.

**c) Hydrodynamic resistance to drainage of TLF**

Inhibition of bubble coalescence explained by the above mentioned causes can also be observed as due to change in interface properties of the TLFs in the film's length scale. Modeling this at the film's length scale amounts to modification of the boundary condition at the gas-liquid interface, because the stability of the TLF depends on the resistance to drainage of the same between two gas-liquid interfaces. The case of a Poisuelle flow in the film results from hydrodynamic resistance at the interface, retarding the drainage rate, thereby increasing stability [17]. On the other hand, a "plug" flow with free slip at the interfaces results in quick drainage as in the case of distilled water [53]. The presence of salts increases the hydrodynamic resistance




(a) Steps involved in coalescence of bubbles

(b) Drainage model of TLF.

Figure 1: (a) Coalescence of approaching bubbles and (b) effective boundary conditions for of a KCl solution between two bubbles. Solid and dashed lines show predictions with no-slip and slip boundary, respectively.[6]

decreasing the mobility of the gas-liquid interface, essentially moving towards the Poisuelle flow regime.

The mobility of the gas-liquid interface is studied at film scale experimentally in [7]. The drainage of liquid film between a solid and a KCl solution was studied by [21] and this study provides evidence that the presence of ions cause viscous drainage (Poisuelle flow) in the thin liquid film. This result can be seen in figure 1b pointing at the possibility that the effect of dissolved salts can be accounted for by modifying the interface rheology of the TLFs. A simulation of bubbles in salt solutions may be then performed by introducing finite slip boundary conditions at gas-liquid interface bounding the TLFs in order to account for the coalescence-inhibiting effects of salts. The inhibition of coalescence due to surfactants can also be modelled in a similar manner by modifying the interface rheology of the TLF.

**d) Choice of numerical method: Smoothed Particle Hydrodynamics (SPH)**
In order to apply a local interface model for approaching bubbles, the choice of a suitable computational fluid dynamics tools from a large number of options needs to be justified. Mesh based CFD tools are known to have difficulties with proper mass conservation at the interface and application of interface boundary conditions within the domain would be cumbersome. The CFD method should also have accurate advection built-in in order to accurately simulate species transport through scalar transport equations. A meshless Lagrangian fluid flow simulation tool would therefore be a viable candidate. Considering these factors, we propose the use of Smoothed Particle Hydrodynamics (SPH) with relevant improvements for phenomena at different scales as a suitable method for the purpose. Being a meshless method with interpolation points also representing constant mass, fluid interfaces are naturally modelled in this method. The SPH method was first introduced in 1977 to test the fission hypothesis [64] in astrophysical systems, and was soon applied to solve hydrodynamic problems since 1985 [62]. In 1994, Monaghan demonstrated SPH's advantage as a tool to study free surface phenomena [57]. Ever since its inception, SPH has remained as the method with the longest continuous development history among meshless methods [23, 24, 26, 32, 46, 49, 65].

A number of improvements to SPH during the past couple of decades now enables the method to account for high density ratios [8], complex geometries and surface tension. A version of SPH, known as incompressible SPH (SPH) which solves for pressure on the particle system is known to be accurate and comparable to traditional CFD methods such as the finite volume method [12]. While free surface modelling was straightforward in the widely used Weakly Com-

pressible SPH model (WCSPH), where a state equation is used to close the Navier Stokes system of equations, an involved approach was needed for simulating fluids in the truly incompressible sense [14, 27]. This issues was solved by using a semi-analytic approach in [5]. The techniques used to simulate incompressibility in liquids are directly linked to the fluctuations in pressure field and are important for accurate simulations [11, 23, 25, 40, 46]. A strictly volume conserving SPH scheme that increases accuracy of SPH simulations in both free surface and internal flow contexts was introduced in [3]. Two more advances in SPH make it further suitable to this project:

- The possibility to couple across scales using a hybrid approach. In [29] a reaction diffision system is solved where local preceipitation of a species is simulated at small scales and coupled to large scale porous media flow.
- The possibility of superposing interparticle forces and relating it to interface properties. In [1, 3] surface tension is related to interparticle forces in a consistent manner. Also, Wiener processes that introduce diffusion at small scales can be modelled in SPH [9], making it an efficient multiscale simulation tool.
- The ease of incorporating more governing equations such as reaction diffusion equation and species transport equation [16, 22] into the meshless domain.

**e) Interface models in SPH**

For simulating effects of surface tension in SPH, multiple approaches can be employed. These approaches can be broadly classified into three. The first approach uses the Continuum Surface Force (CSF) model introduced by Brackbill in [59]. Implementation of this model which uses a 'color' function to obtain gradients and thereby the normals and curvature of the interface was introduced in SPH in [45]. The second approach is to parametrize the free surface explicitly and apply the surface tension force as a volumetric force [19]. A third approach that mimics the molecular basis of surface tension by introducing pair wise forces between particles is found to be more robust in terms of applying it to problems involving multiple phases [3, 37, 44]. Based on the method used by Maxwell to obtain stresses from a molecular system [10], the macroscopic quantities like pressure and surface tension can be related to interparticle forces [3]. Such a pariwise model can be use to apply local local rheological properties to interfaces similar to the interface conditions used in [30, 31].

A number of high performance SPH codes exist as open source and as commercial packages [4, 34]. We have developed an in house SPH code (in development since 2013) that has been validated for a number of three dimensional multiphase flow problems [2]. We plan to further improve and apply this in-house code this project.

## 1.2 Preliminary work

We have improved the traditional Incompressible Smoothed Particle Hydrodynamics (ISPH) algorithm to handle interfaces by superimposing pairwise potentials and relating it to macroscopic interface force coefficients. This allows for local interface boundary conditions in multiphase flows. The algorithm is also implemented in a parallel high performance C code. We have introduced a number of novel features to the Incompressible Smoothed Particle Hydrodynamics algorithm that make it feasible to implement complex interface models. Some of these novel features are:

- An improved free surface algorithm that semi-analytically applies Dirichlet boundary condition at free surface and two phase interfaces where only one phase is solved for, making breakup of domain and other intertial interfacial phenomena to be simulated more accurately and robustly [4]. This also allows for pressure gradients tangential to the free

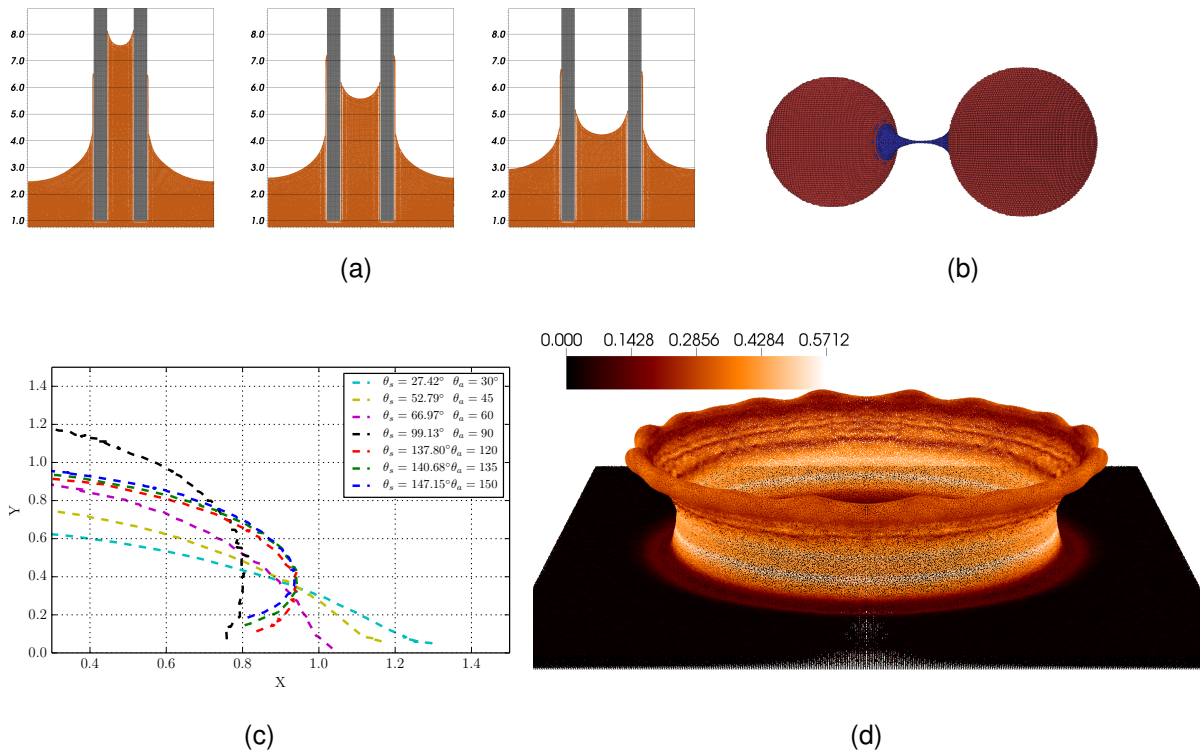



Figure 2: The pairwise force interface model: (a) Simulation of 2D capillary rise for different pore diameters (colored by velocity magnitude.)(b) Simulation of a dynamic liquid bridge rupture as two rigid wet spheres collide and separate [2](c) Simulated contact interface with different apparent contact angles. (d) 3D simulation of splash crown following impact of drop on a film of liquid, the simulation captures splash wavelength realistically [1].

surface close to the free surface.

- Non uniform interfacial tension can be modelled easily.
- A volume conserving pressure equation is used, which ensures incompressibility is better satisfied, without compromising on efficiency [3, 5].
- A surface tension model that is inspired by the molecular theory of capillarity is implemented and validated.
- The model also accounts for three phase contact lines, contact angle dynamics, allowing complex shaped solid interfaces to be accurately modelled.
- The code can also handle rigid and primitively defined bodies [1], such as bubbles with 6 degrees of freedom.
- Species concentration equation and reaction diffusion equation can be added to the set of governing equations without major modifications. framework.

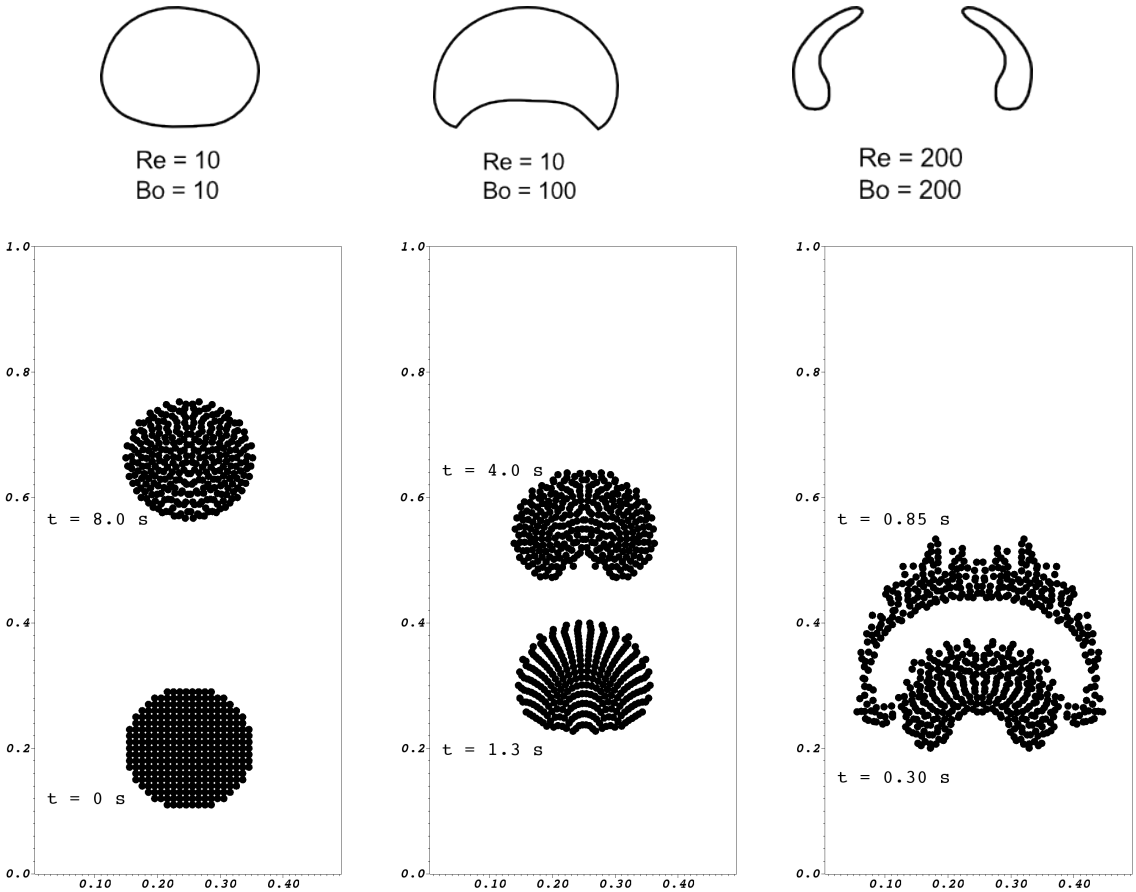
**a) Surface tension and Contact angles**

We have performed a number of validation cases to ensure accuracy of the current code.

Figure 2a shows different simulation results from the implementation of pair-wise force model for surface tension for various interfacial flow phenomena. In Fig. 2a, the capillary rise is validated against theoretical results. In Fig. 2b, a challenging simulation of dynamic rupture of a liquid bridge between colliding solids is presented. The three phase contact angle, which

is a macroscopic parameter is recovered with good accuracy from the pair-wise force model (see Fig. 2c). Contact angles are also relevant for the modeling of thin film as the approching bubbles maintain a certain contact angle depending on the interaction between the air-liquid interfaces. A shared memory parallel simulation of splash of a droplet on thin liquid film is show in Fig. 2d. This recent result shows accurate prediction of the instability that leads to a “crown” formation following splash. The number of waves on the crown corresponds to experimental observation for the same film Weber number. The use of particle method also enables study of material migration and local unsteady phenomena.

**b) Simulation of bubbles using SPH**




(a) $Re = 10,\ Bo = 10$ (b) $Re = 10,\ Bo = 100$ (c) $Re = 200,\ Bo = 200$

Figure 3: SPH simulation of rise of single bubble in a column–stable to unstable regime.

The SPH program discussed above has been validated for single bubble dynamics in 2 and 3 dimensions and for bubble/drop pairs in 3 dimensions. Figure 3 shows the comparison of bubble profiles obtained from a 2D finite volume method simulation which uses a sharp interface technique for the interface with the profile of 2D bubbles simulated using SPH. Though the exact profile has differences the bubbles stability behavious at different Bond and Reynolds numbers match very well. In Fig. 3a and 3b, the bubble is rounded and develops a cusp respectively as seen in the finite volume result. In Fig. 3c, an unstable bubble where inertial forces dominates the viscous and surface tension forces is shown. the corresponding high order of accuracy finite volume simulations are shown in Fig. 3d. Three dimensional simulation of a bubble is shown in Fig. 4. The simulation predicts the vortical flow about the 3D bubble as it rises due

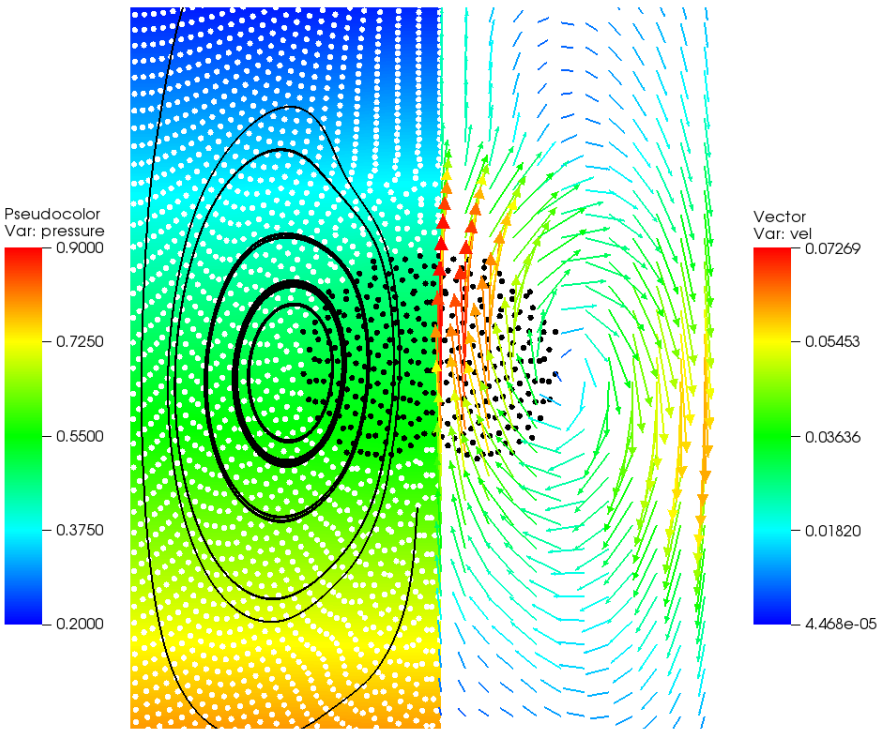
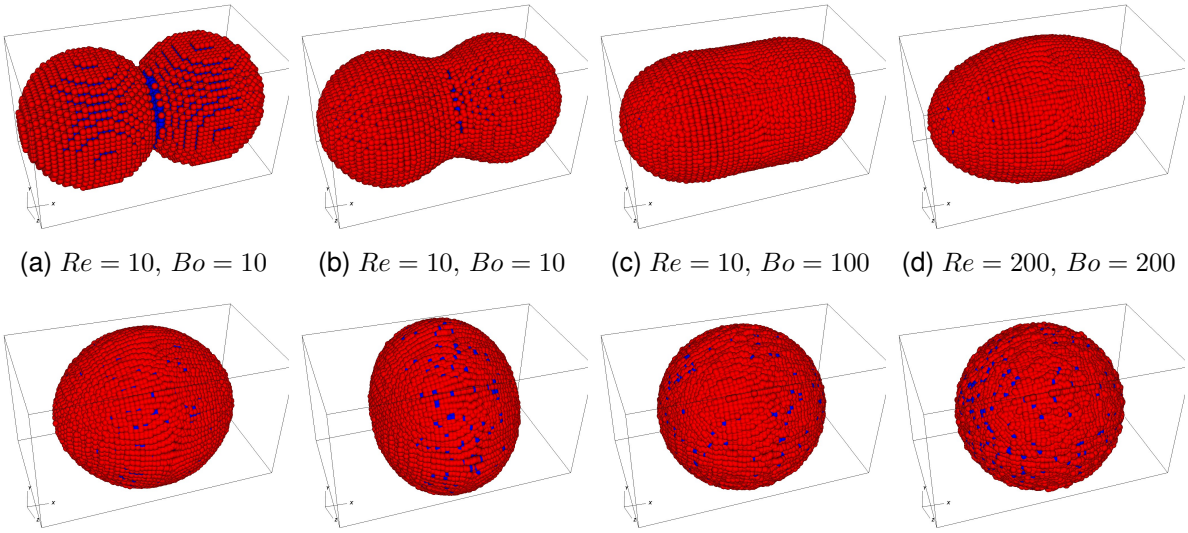



Figure 4: Rise of a 3D bubble—pressure and velocity vectors



(a) $Re = 10,\ Bo = 10$ (b) $Re = 10,\ Bo = 10$ (c) $Re = 10,\ Bo = 100$ (d) $Re = 200,\ Bo = 200$

(e) $Re = 200,\ Bo = 200$ (f) $Re = 200,\ Bo = 200$ (g) $Re = 200,\ Bo = 200$ (h) $Re = 200,\ Bo = 200$

Figure 5: SPH simulation of collapse of bubble in a column

to buoyancy. On the left side, the hydrostatic pressure gradient and the pressure jump across the interface are clearly seen. On the right had side of the image, the arrows show the upward moving vortical flow as the bubble rises upwards, causing the cusp at the bottom of the bubble giving it a concave dimple. The white dot represents the particles belonging to the ambient fluid and the black dot represents the particles belonging to the drop respectively.

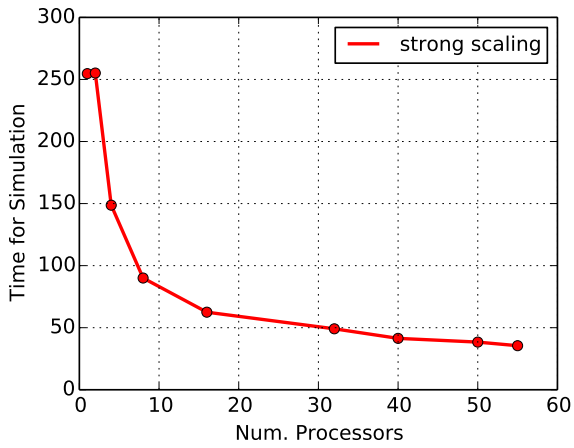



Figure 6: Speedup of the ISPH code: With shared memory parallelization linear speedup is difficult to achieve due to non-localization of data in memory

Figure 5 demonstrates the ability of the SPH program to simulate pair of bubbles coalescing under the influence of surface tension. The red colored particles are particles identified as belonging to the interface and the underlying blue particles represent the interior of the bubbles.

The SPH code uses the shared memory parallelism to distribute computation load among multiple processors that have access to the same memory location. It uses the parallelization library called OpenMP [50] to achieve achieve this [2]. Figure 6 shows the reduction in time with increase in number of processors, achieved with the current code. The algorithm employs a pressure solver which requires creation of large sparse matrices (typically of the order of 0.5 billion non zero numbers for 1 million fluid particles) at every time step. The unordered sparse system is solved using BiCGSTAB [56] algorithm. The creation of this matrix at every time step is the cause for the sub-linear speed up. This will be mitigated by a distributed memory strategy during the course of the proposed project.

## 1.3 List of project-related publications

### 1.3.1 Articles which at the time of proposal submission have been published or officially accepted by publication outlets with scientific quality assurance, listed in standard format; and book publications


[1] P. Nair and G. Tomar. “A study of energy transfer during water entry of solids using incompressible SPH simulations”. In: *Sādhanā* 42.4 (2017), pp. 517–531.

[2] Nair, Prapanch and Pöschel, Thorsten. “Structural changes in wet granular matter due to drainage”. In: *EPJ Web Conf.* 140 (2017), p. 09005.

[3] P. Nair and G. Tomar. “Volume conservation issues in incompressible smoothed particle hydrodynamics”. In: *J Comp. Phys.* 297 (2015), pp. 689–699.

[4] P. Nair and G. Tomar. “An improved free surface modeling for incompressible SPH”. In: *Comput. & Fluids* 102 (2014), pp. 304–314.

[5] P. Nair, A. Vijayakumar, and G. Tomar. “A Deformation Gradient based formulation for Incompressible Smoothed Particle Hydrodynamics”. In: *Proc. 14th Asian Congress of Fluid Mechanics, Hanoi, Vietnam*. Oct. 2013.


### 1.3.2 Other publications


[1] P. Nair and T. Pöschel. “Dynamic capillary phenomena using Incompressible SPH”. In: *Submitted to Chem. Eng. Sci.* (2017).

[2] P. Nair. “Modeling Free Surface Flows and Fluid Structure Interactions using Smoothed Particle Hydrodynamics”. PhD thesis. Bangalore: Department of Mechanical Engineering, Indian Institute of Science, 2015.


### 1.3.3 Patents

– not applicable

# 2 Objectives and work programme

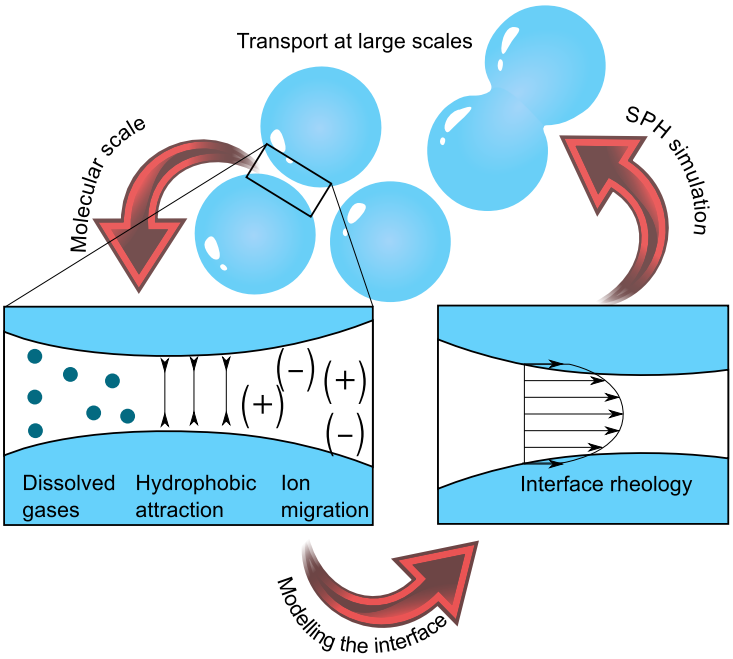



Figure 7: Plan of the proposed project

## 2.1 Anticipated total duration of the project

- 3 years

## 2.2 Objectives

The aim of the project is to functionally relate size of bubbles observed in the bulk of the flow to dissolved constituents in a liquid by simulations that couple the scale at which interface rheology decides coalescence of bubbles to the scale of the bubbles where inertial hydrodynamic effects become important. The project would involve development of an interface model for the thin liquid film to be implemented through pairwise potential forces in a meshless method based on the Smoothed Particle Hydrodynamics and its coupling to SPH simulations in a larger scale. Both the development of the coupled numerical model and the simulations at large scales will be compared with available results in literature during the two different phases of the project. Simple experiments in collaboration with other research institutes may be part of the project due to unavailability of existing research explicitly in this direction.

The mechanism behind inhibition of coalescence of bubbles in the presence of certain salts beyond a critical concentration is yet to be fully understood. However, a number of models exist to describe the stability of thin films and these compare well with experimental observations. To our knowledge these models have not been yet implemented in a numerical simulation of two phase flow. We aim to bridge this research gap and arrive at a simulation tool which responds to salt and surfactant concentration as an input. The Smoothed Particle Hydrodynamics program that we have developed is poised to handle complex boundary conditions while satisfying mass conservation at the interfaces naturally. Given the high performance capability of this simulation

program, simulations with large number of hydrodynamically resolved bubbles is possible as a straightforward extension. The overal scientific plan of coupling the different scales of simulation in the proposal is schematically described through an illustration for clarity in Fig. 7. We present the breakup of the proposed project into Work Packages (WP).

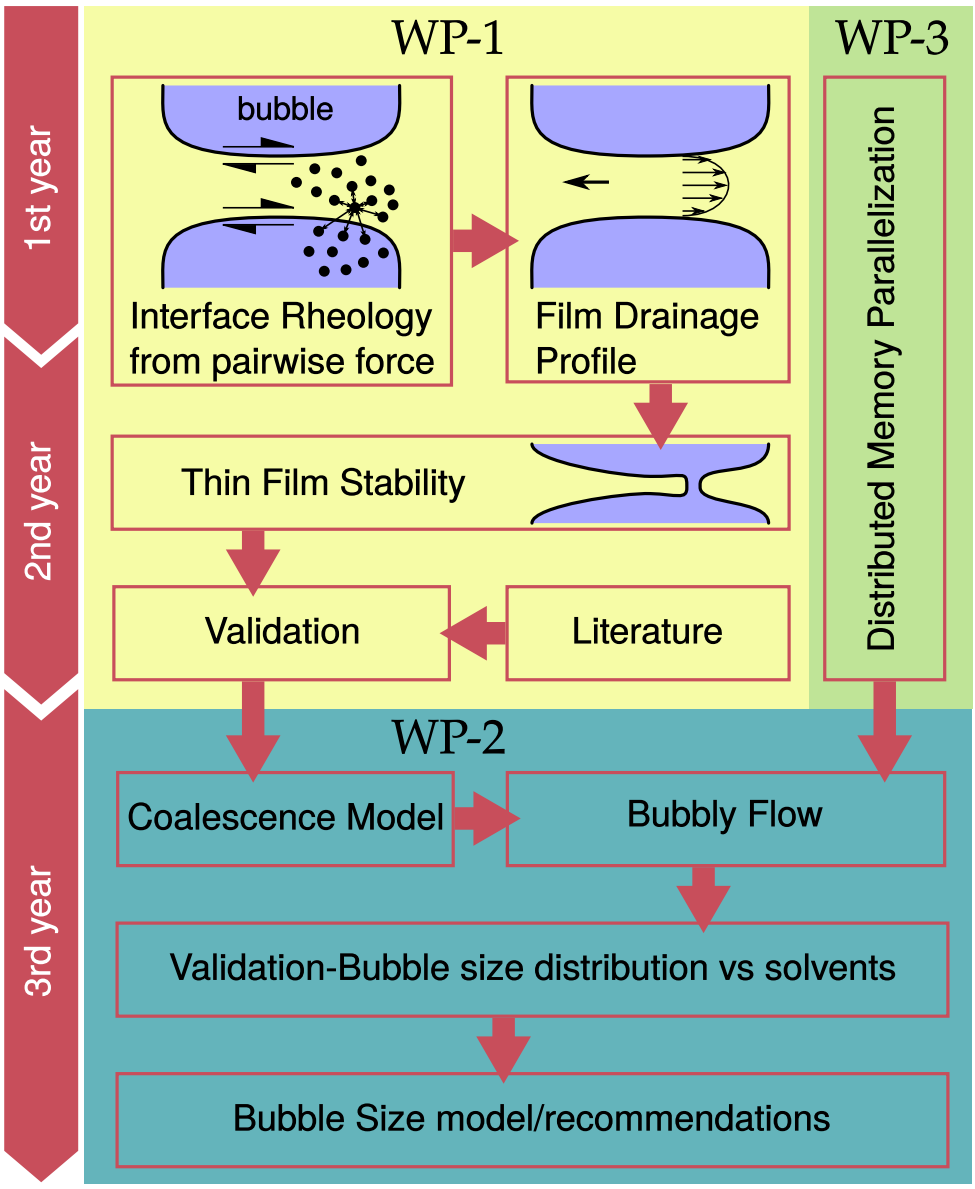



Figure 8: The work packages and their timeline

### 2.3 Work programme including proposed research methods

Our plan for achieving the aforementioned objectives can be broadly divided into three work packages as shown in Fig. 8 and is described below.

**a) Work package # 1 (WP1)**
**Relating thin film models with hydrodynamic boundary conditions**
**Duration:** 2 years
Successful model of the inhibition of coalescence would involve proper modelling of the drainage

between two approaching bubbles. The modeling and validation of drainage between two approaching bubbles in the presence of salinity will be the focus of the first work package WP1. An accurate numerical model for film drainage should consider the following factors:

- Approach speed of the bubbles. The drainage rate is inversely proportional to the radius of film [48, 61, 66]. Therefore increasing the film radius would decrease drainage, leading to a stabilizing effect. For DI water, the critical velocity of approach for films is given by [20] as $u_c = (4/3)^3 \Delta\sigma^2/(\eta\sigma)$ where $\Delta\sigma$ is the surface tension ($\sigma$) difference required to immobilize the interface, and $\eta$ is the viscosity of fluid.
- Surface mobility – the variation of hydrodynamic resistance at the gas-liquid interface ranging from a "plug" flow regime for highly mobile and therefore unstable film to a Poisuelle flow regime for immobile interface representing a stable film.
- Ion concentration – the transition concentration of the salt beyond which thin films become stable depending on approach speed.
- Hydrophobic attraction – The film drainage rate V, based on classical Stefan-Reynolds theory [43] is given as

$$V = -\frac{dh}{dt} = -\frac{2h^3}{3\eta R_f^2}\left(\frac{2\sigma}{R_c} - \Pi\right), \tag{0.3}$$

  where $h$ is the film thickness, $R_f$ and $R_c$ are the radii of the film and the bubble respectively and $\Pi$ is the disjoining pressure of the surface forces. Notably, the van der Waals attraction is are weak and the corresponding Hamacker constant is not significantly influenced by the properties of the ions. The electric double Layer force normally outweighs the van der Waals force, but becomes small in the presence of salts. The attractive hydrophobic forces between the hydrophobic gas-liquid interface becomes significant and is also related to salt concentration.
- Ion specific effects – Effect of different ion pairs with different polarizing effects and affinity towards interface.
- Surface tension gradients – The Gibbs-Marangoni stress due to surface tension gradient with respect to salt concentration retards film thinning by causing radially inward flow in the thin films, and needs to be accounted for.

The interface model can be in the form of interaction potentials between particles or stresses whose divergence can be computed in the SPH particle domain and applied as individual pairwise forces. The SPH approximations allow for the easy converstion of gradients of field variables to discrete forces experience by a particle from other particles in a finite neighborhood. This property is already expounded in our implementation of surface tension model. Also interfaces between different phases can also be heuristically treated through pairwise forces between particles of different phases. The success of simulation of coalescence of two bubbles at different approach speeds and salt concentration can be validated against experimental results on stability of thin films between bubbles, for example as achievd in [6].

An alternative to using SPH to simulate the thin liquid film between approaching bubbles is to use thin film equations using special numerical packages such as CVODE[54] which are specialized in solving stiff equations arising from a large variation in length and time scales in the solution to the thin film equation given in eq. 0.3, for example. Parallelization of this ordinary differential equation is non-trivial and the speed up can be achieved only by spawning separate compute instances when the thin film equation is invoked.

**b) Work package # 2 (WP2)**
**Hydrodynamics of bubbly flows controlled by salinity**
**Duration:** 1 years

Realistic bubbly flows involve bubbles approaching with different speeds. For complex geometries encountered in industrial flow scenarios and for the expected rate of reaction, the size distribution of bubbles and hence the total surface area of bubbles play an important role. The success of WP1 would enable quantitative measure of affinity for bubbles to coalesce as a function of salinity and approach speeds. With this interface boundary condition model, large scale simulations involving hundreds of bubbles can be simulated. The hydrodynamics of each bubble can be accounted for, making it a direct numerical simulation. Scalar transport equation can be solved in the liquid and the variation in transport properties against varying degrees of coalescence can be studied. Since this proposal is part of the second phase of a Priority Program where a number of simulations were performed for bubbly flows by other research groups using Eulerian and Eulerian-Lagrangian simulation method, the results of these can be used for comparisons and recommendations for industrial scale simulations. Once transport properties properties for bubble systems are investigated, available reaction models can be employed to study the effect of salinity. Since SPH is a Lagrangian method, individual species migration can also be studied accurately simply by following material points.

Large scale simulations involving hundreds of bubbles would require simulation domain with tens of millions of mesh points. This would naturally demand distributed memory parallelization of the code.

**c) Work package # 3 (WP3)**
**High performance capability improvements**
**Duration:** 2 years
The coupling of interface boundary conditions would introduce local computational cost spikes in the flow domain. To take account of this and to enable handling large number of bubbles, distributed memory parallelization becomes important. The parallel implementation and speed up improvements of the code will be carried out through the period of WP 1. This would involve help from collaborators as well. The new MPI 3.0 introduced this year enables seamless switch between shared and distributed memory parallelism with little change for different parallel architectures. We aim to implement MPI 3.0 in order to take advantage of modern heterogeneous parallel computers ( for example clusters with Intel Xeon and Xeon Phi nodes available to the applicant at the computing center of the hosting university ).

| Year | 1 | | | | 2 | | | | 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Quarter | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 |
| WP1: Thin film BCs | | | | | | | | | | | | |
| WP2: Flow simulations | | | | | | | | | | | | |
| WP3: HPC improvements | | | | | | | | | | | | |

Table 0.1: The timeline planned for different work packages in the project

At the end of this work package, answering the following specific questions will be explicitly attempted:

- For a known solute causing impedence to bubble coalescence, what is the functional relation between mean bubble size and concentration of the solute?
- What role does hydrodynamic flow conditions play in affecting this mean bubble size?
- Is there a difference in bubble size distribution when different models are used to describe coalescence, such as the pairwise force model and the film drainage equation from lubrication theory.
- How far from the gas inlet does the size distribution stabilize due to coalescence properties?

The answer to these questions could improve bubble column reactors by introducing solutes as a means to control bubble size. Design improvement proposals could be a plausible and applicable outcome of this project.

### 2.4 Data Handling

The primary way to disseminate the results of our research will be through scientific publications in refereed journals. The computer code(s) we use for the simulations are currently version controlled through the version control software, GIT. This makes it easy for publishing the code as an open source project through websites such as github or bitbucket. The results will be published with data sets and code versions, for reproducability of results.

## 3 Bibliography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# 4 Requested modules/funds

## 4.1 Funding for staff

With this proposal, the applicant requests funding for his own position (Temporary Position for Principal Investigator). Given on his experiences in the field of research obtained during his PhD research at the Indian Institute of Science in Bangalore and during his postdoctoral studies at the Institute for Multiscale Simulation at the University Erlangen-Nuremberg, the applicant is well prepared for performing the proposed research project.

The postdoctoral researcher, shall be supported by a student helper (WiHi with bachelor degree). The student helper will support the researcher with performing routine simulations, analyze the simulation results and care for Data Handling (see. 2.4).

Therefore, we request the following funding for stuff:

**Temporary Position for Principal Investigator**

– Dr. Prapanch Nair (TV-L E13 Stufe 2 bis E14 Stufe 1), 100%

- duration: 36 months
- requested funds: 36 months $\times$ 5 700 € (per month) = 205 200 €

**Student helper (WiHi)**

- N.N. 40 hours/month (25%): $1106,25$€ per month)
- duration: 36 months
- requested funds: 36 months $\times$ $1106,25$ € (per month) = $39\,825,00$ €

### 4.2 Funding for direct project costs

**a) Software and consumables**

| | | |
|---|---|---|
| 1. | data storage (hard discs), $1\,000$ € p.a. | $3\,000$€ |
| | total | $3\,000$€ |

**b) Travel**

| | | |
|---|---|---|
| 1. | 2 conferences oversees, each one 1200 € transport, 500 € hotel, 600 € conference fee | 4600€ |
| 2. | 2 conferences in Germany and Europe, each one 700 € transport and hotel, 300 € conference fee | 2000€ |
| | Summe | 6600€ |

**c) Project-related publication expenses**
750€ p.a. = 2250€ for 3 years

## 5 Project requirement

### 5.1 Employment status information

The applicant, Dr. Prapanch Nair, is currently employed as a postdoctoral fellow at the Institute for Multiscale Simulation at the Friedrich Alexander Universit"at Erlangen-Nuremberg, Department for Biological and Chemical Engineering (CBI). The current contract ends October 31, 2018.

### 5.2 First time proposal data

The PI of the current proposal, Dr. Prapanch Nair, is a first-time applicant.

### 5.3 Composition of the project group

The proposed research will be performed at the Institute for Multiscale Simulation. With the following colleagues from this institute exist collaborations or concrete plans for collaboration in relation with the current proposal:

- Prof. Dr. T. Pöschel (head of the institute)
- Prof. Dr. M. Engel (particle simulation)

- Dr. P. Müller - permanent staff member (particle-fluid coupling)
- Dr. Daniel Nasato - DFG-SFB-funded (DEM simulation)
- Dr. Christopher Robert Windows-Yule - non-permanent staff member (Statistical mechanics of particle systems)
- Dr. Mubashir Hussain - non-permanent staff member (particle based fluid simulations)
- Dipl.-Ing. Sebastian Mühlbauer - DFG-funded (chemical reaction kinetics)

## 5.4 Cooperation with other researchers

### 5.4.1 Researcher with existing cooperation agreement for the present project

Apart from the researchers from the Institute for Multiscale Simulations, there is ongoing collaboration with

- Prof. Dr. J. Harting, FAU Erlangen, Institute for Complex Fluids and Interfaces
- Prof. Dr. U. Rüde, FAU Erlangen, Institute for Computer Science

### 5.4.2 Researcher with whom collaboration was done in the past three years (only fields related to the current research project)

Prof. N. Brilliantov (Univ. Leicester); Prof. A. Formella (Univ. Vigo); Prof. J. Harting (Univ. Erlangen); Prof. J. Gallas (Univ. Joao Pessoa); Prof. T. Pöschel (Univ. Erlangen); Prof. G. Tomar (IISc Bangalore); A. Vijayakumar (Univ. Amsterdam);

## 5.5 Scientific equipment

The Institute for Multiscale Simulation has a medium size Linux cluster with about 800 cores which can be used for simulations. Larger simulations shall be performed at the computer center of the university (RRZE-Regionales Rechenzentrum Nürnberg) where the institute has a contingent of computer time.

## 5.6 Project-relevant cooperation in commercial enterprises

not applicable

## 5.7 Project-relevant participation in commercial enterprises

not applicable

# 6 Additional information

- No funding request for this project has been previously submitted to any third party. In case such a request is submitted, the German Research Foundation will be immediately informed.

– The DFG Liaison Officer has been informed of this application